VIT-AP UNIVERSITY

C MINI PROJECT · INTRODUCTION TO PROGRAMMING

LuckyPaws

Animal Rescue and Pet Care System

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Course	Introduction to Programming
Slot	F2+TF2
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About the project

LuckyPaws is a menu-driven C program that helps stray and injured animals get help, lets owners keep their pets' health records with reminders, with a food and care guide and quick answers to common questions. All data is saved in a binary file, so it is still there the next time the program runs.

The program is explained below in nine sections. Each section shows its part of the source code (with the original line numbers of **luckypaws.c**) followed by the output of that part, captured in the Windows Command Prompt.

How to compile and run (Windows)

```
gcc -std=c99 -Wall -Wextra luckypaws.c -o luckypaws.exe
luckypaws.exe
```

Only <stdio.h> and <stdlib.h> are used. The program compiles with no errors and no warnings.

Sections

No.	Section	Lines	Outputs
01	Program setup and main menu	1–145, 1172–1235	1
02	String functions and input checking	146–379	1
03	Date logic	380–480	1
04	Dynamic arrays and recursive search	481–542	1
05	File handling and sample data	543–704	1
06	Display and sorting	705–773	2
07	Rescue & Stray	774–910	4
08	My Pet	911–1086	2
09	Food & care, Ask LuckyPaws and export	1087–1171	3

```c
/*
 * LuckyPaws - Animal Rescue and Pet Care System
 * Name   : GOLTHI SAI LAKSHMI DHEERAJA
 * Reg No : 25BEV7035
 * Course : Introduction to Programming (F2+TF2)
 */

#include <stdio.h>
#include <stdlib.h>

#define DATA_FILE     "luckypaws.dat"
#define REPORT_FILE   "luckypaws_report.txt"
#define MAX_RECORDS   10      /* health records kept per pet */

/* report types and status */
#define STRAY    1
#define INJURED  2
#define MISSING  3
#define OPEN     0
#define RESCUED  1
#define FOUND    2

typedef struct {
    int d, m, y;               /* y == 0 means "none" */
} Date;

/* details that depend on the report type - only one is used at a time */
typedef union {
    int  count;                /* stray   : how many animals */
    char severity[10];         /* injured : Urgent / Serious / Minor */
    char petName[30];          /* missing : the pet's name */
} ReportExtra;

typedef struct {
    int  id;
    int  type;                 /* STRAY / INJURED / MISSING */
    char species[8];           /* Dog / Cat / Other */
    char area[30];
    char city[30];
    char landmark[40];
    Date date;
    char details[80];
    ReportExtra extra;
    char reporter[30];
    char phone[11];
    int  status;               /* OPEN / RESCUED / FOUND */
    char rescuer[40];
    char outcome[30];
} Report;

typedef struct {
    Date date;
    char kind[14];             /* Vaccination / Deworming / Check-up / Treatment */
    char detail[40];
    Date next;                 /* next due date, y == 0 if none */
} HealthRecord;

typedef struct {
    int  id;
    char name[20];
    char species[8];
    char breed[24];
    char sex;                  /* M / F */
    Date dob;
    float weight;              /* kg */
    char colour[20];
    char microchip[16];        /* 15 digits, "" if none */
    int  recordCount;
    HealthRecord records[MAX_RECORDS];   /* array of structures inside a structure */
} Pet;

typedef struct {
    const char *stage;
    const char *meals;
    const char *food;
    const char *care;
} CareGuide;

typedef struct {
```

```
    const char *keys[7];       /* NULL-terminated list of keywords */
    const char *answer;
} Answer;

const char *TYPE_NAMES[4]   = {"", "Stray", "Injured", "Missing"};
const char *STATUS_NAMES[3] = {"Open", "Rescued", "Found"};

/* [species: 0 dog, 1 cat][life stage] */
const CareGuide CARE[2][3] = {
    {
        {"Puppy (under 1 year)", "3-4 small meals a day (3 meals after 6 months)",
         "Good-quality puppy food; soaked kibble for very young puppies; fresh water always",
         "Vaccination course from 6-8 weeks as the vet advises; deworm every 2-3 weeks until 3 months"},
        {"Adult (1-7 years)", "2 meals a day",
         "Balanced adult dog food; rice with boiled chicken or egg as an occasional home meal",
         "Yearly vaccination boosters; deworm every 3 months; daily walks; tick and flea prevention"},
        {"Senior (8+ years)", "2 smaller meals a day",
         "Senior dog food, lower in calories; soft food if teeth are weak",
         "Vet check-up every 6 months; gentle, shorter walks; watch joints and weight"}
    },
    {
        {"Kitten (under 1 year)", "3-4 small meals a day",
         "Kitten food (wet and dry); fresh water always",
         "Vaccination course from 8 weeks as the vet advises; deworm as advised; litter training"},
        {"Adult (1-10 years)", "2 meals a day",
         "Balanced adult cat food - cats need meat-based protein; some wet food for water",
         "Yearly vaccination boosters; regular grooming; clean the litter box daily"},
        {"Senior (11+ years)", "2-3 smaller meals a day",
         "Senior cat food; plenty of water - older cats dehydrate easily",
         "Vet check-up every 6 months; watch for weight loss; soft, warm resting places"}
    }
};
const char *UNSAFE[2] = {
    "chocolate, grapes and raisins, onion and garlic, xylitol (sugar-free sweets), cooked bones, alcohol",
    "lilies (all parts), onion and garlic, chocolate, grapes and raisins, cow's milk, dog food as a diet"
};

#define NUM_ANSWERS 8
const Answer ANSWERS[NUM_ANSWERS] = {
    {{"emergency", "bleeding", "accident", "poison", "seizure", "hit by", NULL},
     "This sounds like an emergency. Take the animal to the nearest vet right away and keep\n  it calm and
warm. You can also file a report with menu option 2."},
    {{"vaccin", "rabies", "booster", "injection", NULL},
     "Puppies and kittens start vaccinations at about 6-8 weeks, then boosters as the vet\n  advises.
Adults usually need yearly boosters, including anti-rabies. Save them in My Pet."},
    {{"fever", "vomit", "diarrh", "not eating", "sick", NULL},
     "A pet that stops eating, keeps vomiting or is very weak should see a vet soon.\n  Never give human
medicines - many are poisonous to pets."},
    {{"chocolate", "grape", "onion", "garlic", "toxic", "lily", NULL},
     "Never give chocolate, grapes, onion, garlic or xylitol. Lilies are very dangerous\n  for cats. If
eaten, contact a vet immediately. See menu option 7."},
    {{"food", "feed", "diet", "meal", "eat", NULL},
     "Puppies and kittens need 3-4 small meals a day; adults usually 2. Always keep fresh\n  water out.
Menu option 7 has guides by age."},
    {{"stray", "street", "adopt", NULL},
     "Report a stray with menu option 1, giving the area and a landmark. To take care of\n  one yourself,
use option 4 and record the outcome."},
    {{"lost", "missing", NULL},
     "Report your pet with menu option 3 - a clear description and where it was last seen.\n  A microchip
makes it much easier to return a lost pet."},
    {{"worm", "tick", "flea", NULL},
     "Adult pets are usually dewormed every 3 months. Check for ticks after walks and use\n  a vet-approved
tick and flea treatment."}
};

/* dynamic arrays (file scope: shared by all functions) */
Report *reports;  int reportCount, reportCap, nextReportId = 1;
Pet    *pets;     int petCount,    petCap,    nextPetId    = 1;
Date today;

/* file layout: version | 4 ints | reports | pets */
#define FILE_VERSION 3
#define HEADER_SIZE ((long)(5 * sizeof(int)))

```

. . .

```
/* ----------------------------------------------------------------
 *  Main menu (array of pointers to functions)
```

```
 * ------------------------------------------------------------------ */

void (*const MENU_ACTIONS[10])(void) = {
    NULL, reportStray, reportInjured, reportMissing, rescueAnimal, viewReports,
    myPetMenu, careGuide, askLuckyPaws, exportReport
};

void printMenu(void)
{
    char date[16];
    const Report *r;
    int open = 0;

    for (r = reports; r < reports + reportCount; r++)
        if (r->status == OPEN)
            open++;
    formatDate(today, date);
    printf("\n=====================================================\n");
    printf("     LuckyPaws - Rescue . Care . Connect\n");
    printf("=====================================================\n");
    printf(" Today: %s   Open reports: %d   Pets: %d\n", date, open, petCount);
    printf("-----------------------------------------------------\n");
    printf(" RESCUE & STRAY\n");
    printf("  1. Report a stray animal\n");
    printf("  2. Report an injured animal\n");
    printf("  3. Report a missing pet\n");
    printf("  4. Rescue a stray / mark a pet found\n");
    printf("  5. View reports\n");
    printf(" MY PET & MORE\n");
    printf("  6. My Pet (profiles, health records, reminders)\n");
    printf("  7. Pet food & care guide\n");
    printf("  8. Ask LuckyPaws\n");
    printf("  9. Export summary report\n");
    printf("  0. Save and exit\n");
    printf("-----------------------------------------------------\n");
}

int main(void)
{
    int choice;

    printf("LuckyPaws - Rescue . Care . Connect\n");
    today = readToday();
    if (!loadAll()) {
        loadSampleData();
        saveAll();
        printf("First run: sample reports and pets loaded into %s.\n", DATA_FILE);
    }

    do {
        printMenu();
        choice = readInt("Enter your choice: ", 0, 9);
        if (choice != 0)
            MENU_ACTIONS[choice]();          /* call through the function pointer */
    } while (choice != 0);

    if (saveAll())
        printf("\nData saved to %s. Thank you for caring for animals!\n", DATA_FILE);
    free(reports);
    free(pets);
    return 0;
}
```

```
C:\Users\golth\OneDrive\Documents\LuckyPaws>ren luckypaws.c.txt luckypaws.c

C:\Users\golth\OneDrive\Documents\LuckyPaws>gcc -std=c99 -Wall -Wextra luckypaws.c -o luckypaws.exe

C:\Users\golth\OneDrive\Documents\LuckyPaws>luckypaws.exe
LuckyPaws - Rescue . Care . Connect
Enter today's date (dd/mm/yyyy): 25-10-2026
  ! Invalid date. Use dd/mm/yyyy, e.g. 25/10/2026.
Enter today's date (dd/mm/yyyy): 25/10/2026
First run: sample reports and pets loaded into luckypaws.dat.

================================================
     LuckyPaws - Rescue . Care . Connect
================================================
 Today: 25/10/2026   Open reports: 3   Pets: 3
------------------------------------------------
 RESCUE & STRAY
  1. Report a stray animal
  2. Report an injured animal
  3. Report a missing pet
  4. Rescue a stray / mark a pet found
  5. View reports
 MY PET & MORE
  6. My Pet (profiles, health records, reminders)
  7. Pet food & care guide
  8. Ask LuckyPaws
  9. Export summary report
  0. Save and exit
------------------------------------------------
Enter your choice:
```

Figure 1.1 Compiling with gcc and running luckypaws.exe: an invalid date is rejected, sample data is loaded and the main menu is shown.

```c
/* ---------------------------------------------------------------
 *  Own character and string functions (character arrays + pointers)
 * --------------------------------------------------------------- */

int isDigitChar(char c)
{
    return c >= '0' && c <= '9';
}

int isLetter(char c)
{
    return (c >= 'a' && c <= 'z') || (c >= 'A' && c <= 'Z');
}

int isSpaceChar(char c)
{
    return c == ' ' || c == '\t' || c == '\n' || c == '\r';
}

char lowerCase(char c)
{
    return (c >= 'A' && c <= 'Z') ? (char)(c - 'A' + 'a') : c;
}

char upperCase(char c)
{
    return (c >= 'a' && c <= 'z') ? (char)(c - 'a' + 'A') : c;
}

/* length of a string: end pointer minus start pointer */
int strLength(const char *s)
{
    const char *p = s;

    while (*p != '\0')
        p++;
    return (int)(p - s);
}

void strCopy(char *dest, const char *src)
{
    while ((*dest++ = *src++) != '\0')
        ;
}

/* 0 when equal, otherwise the difference of the first characters that differ */
int strCompare(const char *a, const char *b)
{
    while (*a != '\0' && *a == *b) {
        a++;
        b++;
    }
    return (unsigned char)*a - (unsigned char)*b;
}

/* 1 when the character c appears in s */
int hasChar(const char *s, char c)
{
    for (; *s != '\0'; s++)
        if (*s == c)
            return 1;
    return 0;
}

/* 1 when word appears anywhere inside text */
int containsWord(const char *text, const char *word)
{
    int i, j;

    for (i = 0; text[i] != '\0'; i++) {
        for (j = 0; word[j] != '\0' && text[i + j] == word[j]; j++)
            ;
        if (word[j] == '\0')
            return 1;
    }
    return 0;
}

/* removes spaces at both ends */
```

```
225 void trim(char *s)
226 {
227     int start = 0, i, n;
228 
229     while (s[start] != '\0' && isSpaceChar(s[start]))
230         start++;
231     for (i = 0; s[start + i] != '\0'; i++)      /* shift the text left */
232         s[i] = s[start + i];
233     s[i] = '\0';
234     n = strLength(s);
235     while (n > 0 && isSpaceChar(s[n - 1]))
236         s[--n] = '\0';
237 }
238 
239 int sameText(const char *a, const char *b)
240 {
241     while (*a && *b) {
242         if (lowerCase(*a) != lowerCase(*b))
243             return 0;
244         a++;
245         b++;
246     }
247     return *a == *b;
248 }
249 
250 void toLower(const char *in, char *out)
251 {
252     while (*in)
253         *out++ = lowerCase(*in++);
254     *out = '\0';
255 }
256 
257 /* ---------------------------------------------------------------------
258  *  Input helpers (fgets based)
259  * --------------------------------------------------------------------- */
260 
261 void readLine(const char *prompt, char *buf, int size)
262 {
263     printf("%s", prompt);
264     fflush(stdout);
265     if (fgets(buf, size, stdin) == NULL) {
266         printf("\nInput ended. Exiting.\n");
267         exit(0);
268     }
269     if (!hasChar(buf, '\n')) {
270         int c;
271         while ((c = getchar()) != '\n' && c != EOF)
272             ;
273     }
274     trim(buf);
275 }
276 
277 int readInt(const char *prompt, int min, int max)
278 {
279     char buf[32], extra;
280     int v;
281 
282     for (;;) {
283         readLine(prompt, buf, sizeof buf);
284         /* exactly one number and nothing after it */
285         if (sscanf(buf, "%d%c", &v, &extra) == 1 && v >= min && v <= max)
286             return v;
287         printf("  ! Please enter a number between %d and %d.\n", min, max);
288     }
289 }
290 
291 float readFloat(const char *prompt, float min, float max)
292 {
293     char buf[32], extra;
294     float v;
295 
296     for (;;) {
297         readLine(prompt, buf, sizeof buf);
298         if (sscanf(buf, "%f%c", &v, &extra) == 1 && v >= min && v <= max)
299             return v;
300         printf("  ! Please enter a value between %.1f and %.1f.\n", min, max);
301     }
302 }
303 
```

```
int readYesNo(const char *prompt)
{
    char buf[8];

    for (;;) {
        readLine(prompt, buf, sizeof buf);
        if (buf[0] == 'y' || buf[0] == 'Y')
            return 1;
        if (buf[0] == 'n' || buf[0] == 'N')
            return 0;
        printf("  ! Please answer y or n.\n");
    }
}

/* any text of at least minLen characters (an empty answer is allowed when minLen is 0) */
void readText(const char *prompt, char *buf, int size, int minLen)
{
    for (;;) {
        readLine(prompt, buf, size);
        if ((int)strLength(buf) >= minLen)
            return;
        printf("  ! Please enter at least %d characters.\n", minLen);
    }
}

/* letters, spaces and dots only */
void readWords(const char *prompt, char *buf, int size)
{
    for (;;) {
        int ok, i;

        readLine(prompt, buf, size);
        ok = buf[0] != '\0';
        for (i = 0; buf[i] && ok; i++)
            if (!isLetter(buf[i]) && buf[i] != ' ' && buf[i] != '.')
                ok = 0;
        if (ok)
            return;
        printf("  ! Required. Use letters and spaces only.\n");
    }
}

void readPhone(const char *prompt, char *buf)
{
    char tmp[32];

    for (;;) {
        int ok, i;

        readLine(prompt, tmp, sizeof tmp);
        ok = strLength(tmp) == 10 && tmp[0] >= '6' && tmp[0] <= '9';
        for (i = 0; tmp[i] && ok; i++)
            if (!isDigitChar(tmp[i]))
                ok = 0;
        if (ok) {
            strCopy(buf, tmp);
            return;
        }
        printf("  ! Enter a valid 10-digit mobile number (starts with 6-9).\n");
    }
}

/* Dog / Cat, plus Other when allowOther is set (any letter case) */
void readSpecies(const char *prompt, char *out, int allowOther)
{
    char buf[16];

    for (;;) {
        readLine(prompt, buf, sizeof buf);
        if (sameText(buf, "dog"))                { strCopy(out, "Dog");   return; }
        if (sameText(buf, "cat"))                { strCopy(out, "Cat");   return; }
        if (allowOther && sameText(buf, "other")) { strCopy(out, "Other"); return; }
        printf("  ! Type %s.\n", allowOther ? "Dog, Cat or Other" : "Dog or Cat");
    }
}

```

```
Enter your choice: abc
  ! Please enter a number between 0 and 9.
Enter your choice: 12
  ! Please enter a number between 0 and 9.
Enter your choice: 0

Data saved to luckypaws.dat. Thank you for caring for animals!

C:\Users\golth\OneDrive\Documents\LuckyPaws>
```

Figure 2.1 Wrong menu input: letters and an out-of-range number are rejected; choosing 0 saves the data and exits.

```
/* ---------------------------------------------------------------
 *  Date logic
 * --------------------------------------------------------------- */

int isLeap(int y)
{
    return (y % 4 == 0 && y % 100 != 0) || y % 400 == 0;
}

int isValidDate(Date dt)
{
    const int days[] = {31, 28, 31, 30, 31, 30, 31, 31, 30, 31, 30, 31};

    if (dt.y < 1990 || dt.y > 2100 || dt.m < 1 || dt.m > 12 || dt.d < 1)
        return 0;
    return dt.d <= ((dt.m == 2 && isLeap(dt.y)) ? 29 : days[dt.m - 1]);
}

long dateToDays(Date dt)
{
    int y = dt.y, m = dt.m;

    if (m < 3) {
        y--;
        m += 12;
    }
    return 365L * y + y / 4 - y / 100 + y / 400 + (153 * (m - 3) + 2) / 5 + dt.d;
}

int ageOn(Date dob, Date on)
{
    int age = on.y - dob.y;

    if (on.m < dob.m || (on.m == dob.m && on.d < dob.d))
        age--;
    return age;
}

int compareDates(Date a, Date b)
{
    long diff = dateToDays(a) - dateToDays(b);
    return diff < 0 ? -1 : diff > 0 ? 1 : 0;
}

void formatDate(Date dt, char *out)
{
    if (dt.y == 0)
        strCopy(out, "-");
    else
        sprintf(out, "%02d/%02d/%04d", dt.d, dt.m, dt.y);
}

Date makeDate(int d, int m, int y)
{
    Date dt;

    dt.d = d;
    dt.m = m;
    dt.y = y;
    return dt;
}

/*
 * dd/mm/yyyy. "0" means none when allowNone is set.
 * future = 1 allows future dates only (due dates), 0 allows past dates only.
 */
Date readDate(const char *prompt, int allowNone, int future)
{
    char buf[32];
    Date dt;

    for (;;) {
        readLine(prompt, buf, sizeof buf);
        if (allowNone && strCompare(buf, "0") == 0)
            return makeDate(0, 0, 0);
        if (sscanf(buf, "%d/%d/%d", &dt.d, &dt.m, &dt.y) == 3 && isValidDate(dt)) {
            if (!future && compareDates(dt, today) <= 0)
                return dt;
            if (future && compareDates(dt, today) > 0)
```

```
459                 return dt;
460             printf("  ! The date must be %s.\n", future ? "after today" : "today or earlier");
461         } else {
462             printf("  ! Invalid date. Use dd/mm/yyyy%s.\n", allowNone ? ", or 0 for none" : "");
463         }
464     }
465 }
466
467 /* today's date is entered once when the program starts */
468 Date readToday(void)
469 {
470     char buf[32];
471     Date dt;
472
473     for (;;) {
474         readLine("Enter today's date (dd/mm/yyyy): ", buf, sizeof buf);
475         if (sscanf(buf, "%d/%d/%d", &dt.d, &dt.m, &dt.y) == 3 && isValidDate(dt) && dt.y >= 2024)
476             return dt;
477         printf("  ! Invalid date. Use dd/mm/yyyy, e.g. 25/10/2026.\n");
478     }
479 }
480
```

```
Enter your choice: 6

--- My Pet ---
1. View my pets  2. Pet profile & records  3. Add a pet  4. Add a health record  5. Reminders
Choice: 5

--- Due in the next 30 days (or overdue) ---
  Max      DHPPi + Lepto            05/11/2026  due in 11 days
  Tiger    Anti-rabies vaccine      10/11/2026  due in 16 days
```

Figure 3.1 Reminders: health records due in the next 30 days, worked out from the day counts.

```
481  /* ------------------------------------------------------------------
482   *  Dynamic arrays and recursive search
483   * ------------------------------------------------------------------ */
484
485  void outOfMemory(void)
486  {
487      printf("Out of memory!\n");
488      exit(1);
489  }
490
491  /* each array grows by doubling: malloc the first block, realloc after that */
492  void growReports(void)
493  {
494      Report *bigger;
495
496      if (reportCount < reportCap)
497          return;
498      reportCap = reportCap ? reportCap * 2 : 8;
499      bigger = reports == NULL ? malloc(reportCap * sizeof(Report)) : realloc(reports, reportCap *
sizeof(Report));
500      if (bigger == NULL)
501          outOfMemory();
502      reports = bigger;
503  }
504
505  void growPets(void)
506  {
507      Pet *bigger;
508
509      if (petCount < petCap)
510          return;
511      petCap = petCap ? petCap * 2 : 8;
512      bigger = pets == NULL ? malloc(petCap * sizeof(Pet)) : realloc(pets, petCap * sizeof(Pet));
513      if (bigger == NULL)
514          outOfMemory();
515      pets = bigger;
516  }
517
518  /* recursive binary search over the ids (each list is always sorted by id) */
519  int findReport(int low, int high, int id)
520  {
521      int mid;
522
523      if (low > high)
524          return -1;
525      mid = low + (high - low) / 2;
526      if (reports[mid].id == id)
527          return mid;
528      return reports[mid].id < id ? findReport(mid + 1, high, id) : findReport(low, mid - 1, id);
529  }
530
531  int findPet(int low, int high, int id)
532  {
533      int mid;
534
535      if (low > high)
536          return -1;
537      mid = low + (high - low) / 2;
538      if (pets[mid].id == id)
539          return mid;
540      return pets[mid].id < id ? findPet(mid + 1, high, id) : findPet(low, mid - 1, id);
541  }
542
```

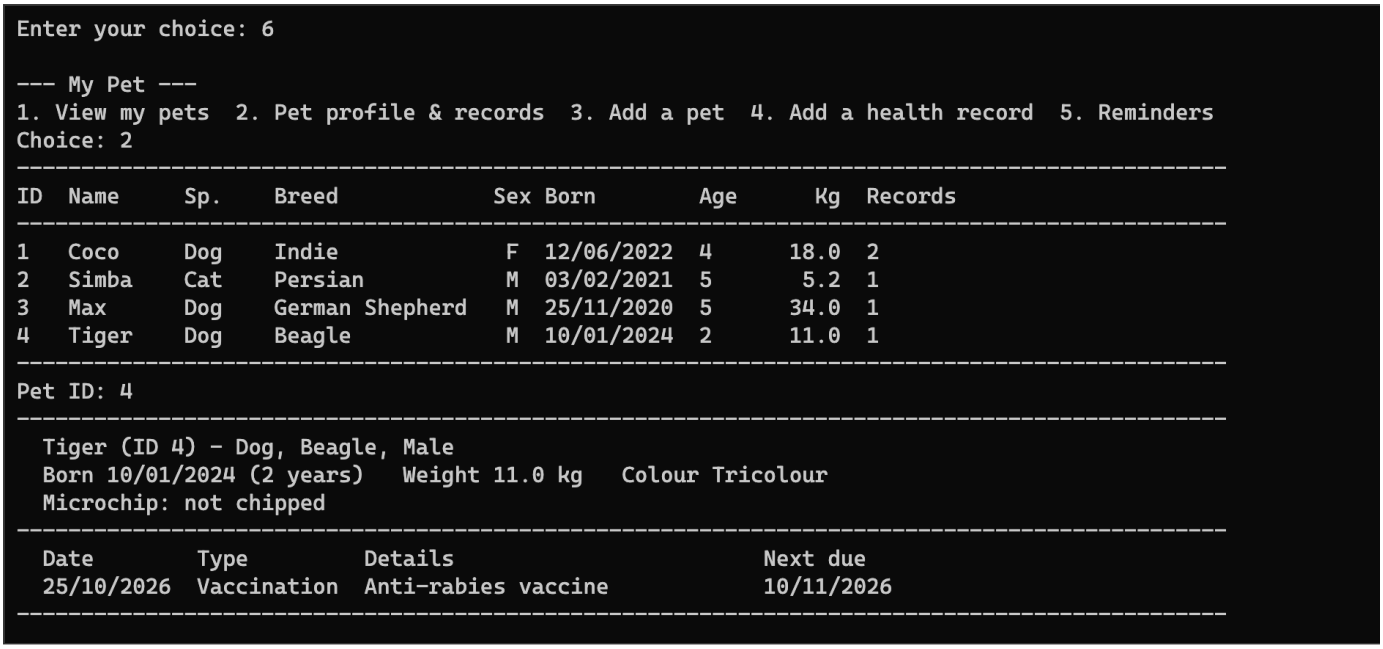


Figure 4.1 Pet profile found by its ID with the recursive binary search.

```c
/* ---------------------------------------------------------------
 *  File handling
 * --------------------------------------------------------------- */

int saveAll(void)
{
    FILE *fp = fopen(DATA_FILE, "wb");
    int header[5];

    if (fp == NULL) {
        printf("  ! Could not save to %s\n", DATA_FILE);
        return 0;
    }
    header[0] = FILE_VERSION;
    header[1] = reportCount; header[2] = petCount;
    header[3] = nextReportId; header[4] = nextPetId;
    fwrite(header, sizeof(int), 5, fp);
    fwrite(reports, sizeof(Report), reportCount, fp);
    fwrite(pets, sizeof(Pet), petCount, fp);
    fclose(fp);
    return 1;
}

/* random access: rewrites one report in place after a rescue */
void saveReport(int idx)
{
    FILE *fp = fopen(DATA_FILE, "r+b");

    if (fp == NULL) {
        saveAll();
        return;
    }
    fseek(fp, HEADER_SIZE + (long)idx * (long)sizeof(Report), SEEK_SET);
    fwrite(&reports[idx], sizeof(Report), 1, fp);
    fclose(fp);
}

int loadAll(void)
{
    FILE *fp = fopen(DATA_FILE, "rb");
    int header[5];

    if (fp == NULL)
        return 0;
    /* the file starts with a version number: a file from an older program is not used */
    if (fread(header, sizeof(int), 5, fp) != 5 || header[0] != FILE_VERSION ||
        header[1] < 0 || header[2] < 0) {
        printf("  ! %s is from an older version or damaged - starting with sample data.\n", DATA_FILE);
        fclose(fp);
        return 0;
    }
    reportCount = header[1]; petCount = header[2];
    nextReportId = header[3]; nextPetId = header[4];
    reportCap = reportCount + 8; petCap = petCount + 8;
    reports = calloc(reportCap, sizeof(Report));
    pets    = calloc(petCap, sizeof(Pet));
    if (reports == NULL || pets == NULL)
        outOfMemory();
    if ((int)fread(reports, sizeof(Report), reportCount, fp) != reportCount ||
        (int)fread(pets, sizeof(Pet), petCount, fp) != petCount) {
        fclose(fp);
        return 0;
    }
    fclose(fp);
    return 1;
}

/* ---------------------------------------------------------------
 *  Sample data for the first run
 * --------------------------------------------------------------- */

Report *newReport(int type, const char *species, const char *area, const char *city,
                  const char *landmark, Date date, const char *details,
                  const char *reporter, const char *phone)
{
    Report *r;
    Report blankReport = {0};

    growReports();
```

```
    r = &reports[reportCount++];
    *r = blankReport;
    r->id = nextReportId++;
    r->type = type;
    strCopy(r->species, species);
    strCopy(r->area, area);
    strCopy(r->city, city);
    strCopy(r->landmark, landmark);
    r->date = date;
    strCopy(r->details, details);
    strCopy(r->reporter, reporter);
    strCopy(r->phone, phone);
    r->status = OPEN;
    return r;
}

Pet *newPet(const char *name, const char *species, const char *breed, char sex,
                  Date dob, float weight, const char *colour, const char *chip)
{
    Pet *p;
    Pet blankPet = {0};

    growPets();
    p = &pets[petCount++];
    *p = blankPet;
    p->id = nextPetId++;
    strCopy(p->name, name);
    strCopy(p->species, species);
    strCopy(p->breed, breed);
    p->sex = sex;
    p->dob = dob;
    p->weight = weight;
    strCopy(p->colour, colour);
    strCopy(p->microchip, chip);
    return p;
}

void addRecord(Pet *p, Date date, const char *kind, const char *detail, Date next)
{
    HealthRecord *h = &p->records[p->recordCount++];

    h->date = date;
    strCopy(h->kind, kind);
    strCopy(h->detail, detail);
    h->next = next;
}

void loadSampleData(void)
{
    Report *r;
    Pet *p;

    r = newReport(INJURED, "Dog", "Benz Circle", "Vijayawada", "Near bus stop", makeDate(24, 10, 2026),
                  "Limping on back left leg, wound on paw", "Kavya", "9876500011");
    strCopy(r->extra.severity, "Urgent");
    r = newReport(STRAY, "Cat", "Brodipet", "Guntur", "4th line market", makeDate(22, 10, 2026),
                  "Mother cat with 3 kittens, looks hungry", "Rahul", "9876500012");
    r->extra.count = 4;
    r = newReport(MISSING, "Dog", "Labbipet", "Vijayawada", "Near park gate", makeDate(20, 10, 2026),
                  "Brown Labrador, red collar", "Sneha", "9876500013");
    strCopy(r->extra.petName, "Bruno");
    r = newReport(STRAY, "Dog", "Thullur Road", "Amaravati", "Petrol bunk", makeDate(18, 10, 2026),
                  "Friendly puppy, about 3 months old", "Arjun", "9876500014");
    r->extra.count = 1;
    r->status = RESCUED;
    strCopy(r->rescuer, "Paws Shelter Amaravati");
    strCopy(r->outcome, "Taken to shelter");
    r = newReport(INJURED, "Cat", "Kothapet", "Guntur", "Behind temple", makeDate(15, 10, 2026),
                  "Eye infection", "Divya", "9876500015");
    strCopy(r->extra.severity, "Serious");
    r->status = RESCUED;
    strCopy(r->rescuer, "Dr. Ramana (vet)");
    strCopy(r->outcome, "Treated at vet clinic");

    p = newPet("Coco", "Dog", "Indie", 'F', makeDate(12, 6, 2022), 18.0f, "Fawn", "");
    addRecord(p, makeDate(10, 8, 2026), "Vaccination", "Anti-rabies booster", makeDate(10, 8, 2027));
    addRecord(p, makeDate(15, 9, 2026), "Deworming", "Tablet given", makeDate(15, 12, 2026));
    p = newPet("Simba", "Cat", "Persian", 'M', makeDate(3, 2, 2021), 5.2f, "White", "985112003456789");
    addRecord(p, makeDate(20, 7, 2026), "Vaccination", "FVRCP", makeDate(20, 7, 2027));
```

```
    p = newPet("Max", "Dog", "German Shepherd", 'M', makeDate(25, 11, 2020), 34.0f, "Black and tan", "");
    addRecord(p, makeDate(1, 3, 2026), "Vaccination", "DHPPi + Lepto", makeDate(5, 11, 2026));
}

```

```
C:\Users\golth\OneDrive\Documents\LuckyPaws>luckypaws.exe
LuckyPaws - Rescue . Care . Connect
Enter today's date (dd/mm/yyyy): 25/10/2026

================================================
     LuckyPaws - Rescue . Care . Connect
================================================
 Today: 25/10/2026   Open reports: 5   Pets: 4
------------------------------------------------
 RESCUE & STRAY
  1. Report a stray animal
  2. Report an injured animal
  3. Report a missing pet
  4. Rescue a stray / mark a pet found
  5. View reports
 MY PET & MORE
  6. My Pet (profiles, health records, reminders)
  7. Pet food & care guide
  8. Ask LuckyPaws
  9. Export summary report
  0. Save and exit
------------------------------------------------
Enter your choice: 5

Show: 1. All  2. Open  3. Stray  4. Injured  5. Missing  6. One city  7. All, by city
Choice: 1

-----------------------------------------------------------------------------------------------
ID  Type     Animal Area           City        Date       Details        Status
-----------------------------------------------------------------------------------------------
6   Injured  Dog    Arundelpet     Guntur      25/10/2026 Urgent         Open
    Hit by a bike, bleeding from the leg
7   Stray    Cat    Kothapet       Guntur      24/10/2026 3 animal(s)    Open
    Three kittens near the bins, hungry
1   Injured  Dog    Benz Circle    Vijayawada  24/10/2026 Urgent         Open
    Limping on back left leg, wound on paw
8   Missing  Dog    MG Road        Vijayawada  23/10/2026 Pet: Tommy     Open
    White Pomeranian, blue collar
2   Stray    Cat    Brodipet       Guntur      22/10/2026 4 animal(s)    Rescued
    Mother cat with 3 kittens, looks hungry  [Guntur Animal Shelter: Taken to shelter]
3   Missing  Dog    Labbipet       Vijayawada  20/10/2026 Pet: Bruno     Open
    Brown Labrador, red collar
4   Stray    Dog    Thullur Road   Amaravati   18/10/2026 1 animal(s)    Rescued
    Friendly puppy, about 3 months old  [Paws Shelter Amaravati: Taken to shelter]
5   Injured  Cat    Kothapet       Guntur      15/10/2026 Serious        Rescued
    Eye infection  [Dr. Ramana (vet): Treated at vet clinic]
-----------------------------------------------------------------------------------------------
8 report(s) shown.
```

Figure 5.1 Running the program again: everything is read back from luckypaws.dat.

```
 705  /* ------------------------------------------------------------------
 706   *  Display helpers
 707   * ------------------------------------------------------------------ */
 708
 709  void printLine(void)
 710  {
 711      printf("-------------------------------------------------------------------------------------------------
\n");
 712  }
 713
 714  void extraText(const Report *r, char *out)
 715  {
 716      switch (r->type) {
 717      case STRAY:   sprintf(out, "%d animal(s)", r->extra.count);    break;
 718      case INJURED: sprintf(out, "%s", r->extra.severity);           break;
 719      default:      sprintf(out, "Pet: %s", r->extra.petName);       break;
 720      }
 721  }
 722
 723  void printReport(const Report *r)
 724  {
 725      char date[16], extra[40];
 726
 727      formatDate(r->date, date);
 728      extraText(r, extra);
 729      printf("%-3d %-8s %-6s %-14.14s %-11.11s %-10s %-14.14s %s\n",
 730             r->id, TYPE_NAMES[r->type], r->species, r->area, r->city, date, extra, STATUS_NAMES[r->status]);
 731      printf("    %s", r->details);
 732      if (r->status != OPEN)
 733          printf("  [%s: %s]", r->rescuer, r->outcome);
 734      printf("\n");
 735  }
 736
 737  void printReportHeader(void)
 738  {
 739      printLine();
 740      printf("%-3s %-8s %-6s %-14s %-11s %-10s %-14s %s\n", "ID", "Type", "Animal", "Area", "City", "Date",
"Details", "Status");
 741      printLine();
 742  }
 743
 744  /* ------------------------------------------------------------------
 745   *  Sorting (the compare rule is passed as a pointer to a function)
 746   * ------------------------------------------------------------------ */
 747
 748  int byDateNewest(const Report *x, const Report *y)
 749  {
 750      int c = compareDates(y->date, x->date);
 751      return c ? c : y->id - x->id;
 752  }
 753
 754  int byCity(const Report *x, const Report *y)
 755  {
 756      int c = strCompare(x->city, y->city);
 757      return c ? c : compareDates(y->date, x->date);
 758  }
 759
 760  /* insertion sort of an array of report pointers, using any compare function */
 761  void sortReports(const Report *view[], int n, int (*compare)(const Report *, const Report *))
 762  {
 763      int i, j;
 764
 765      for (i = 1; i < n; i++) {
 766          const Report *key = view[i];
 767
 768          for (j = i - 1; j >= 0 && compare(view[j], key) > 0; j--)
 769              view[j + 1] = view[j];
 770          view[j + 1] = key;
 771      }
 772  }
 773
```

```
Enter your choice: 5

Show: 1. All  2. Open  3. Stray  4. Injured  5. Missing  6. One city  7. All, by city
Choice: 1

-----------------------------------------------------------------------------------------------
ID  Type     Animal Area           City        Date       Details        Status
-----------------------------------------------------------------------------------------------
1   Injured  Dog    Benz Circle    Vijayawada  24/10/2026 Urgent         Open
    Limping on back left leg, wound on paw
2   Stray    Cat    Brodipet       Guntur      22/10/2026 4 animal(s)    Open
    Mother cat with 3 kittens, looks hungry
3   Missing  Dog    Labbipet       Vijayawada  20/10/2026 Pet: Bruno     Open
    Brown Labrador, red collar
4   Stray    Dog    Thullur Road   Amaravati   18/10/2026 1 animal(s)    Rescued
    Friendly puppy, about 3 months old  [Paws Shelter Amaravati: Taken to shelter]
5   Injured  Cat    Kothapet       Guntur      15/10/2026 Serious        Rescued
    Eye infection  [Dr. Ramana (vet): Treated at vet clinic]
-----------------------------------------------------------------------------------------------
5 report(s) shown.
```

Figure 6.1 All reports, newest first.

```
Enter your choice: 5

Show: 1. All  2. Open  3. Stray  4. Injured  5. Missing  6. One city  7. All, by city
Choice: 6
City: Guntur

-----------------------------------------------------------------------------------------------
ID  Type     Animal Area           City        Date       Details        Status
-----------------------------------------------------------------------------------------------
6   Injured  Dog    Arundelpet     Guntur      25/10/2026 Urgent         Open
    Hit by a bike, bleeding from the leg
7   Stray    Cat    Kothapet       Guntur      24/10/2026 3 animal(s)    Open
    Three kittens near the bins, hungry
2   Stray    Cat    Brodipet       Guntur      22/10/2026 4 animal(s)    Rescued
    Mother cat with 3 kittens, looks hungry  [Guntur Animal Shelter: Taken to shelter]
5   Injured  Cat    Kothapet       Guntur      15/10/2026 Serious        Rescued
    Eye infection  [Dr. Ramana (vet): Treated at vet clinic]
-----------------------------------------------------------------------------------------------
4 report(s) shown.
```

Figure 6.2 Reports of one city (Guntur) only.

```
774 /* ---------------------------------------------------------------
775  *  Rescue & Stray
776  * --------------------------------------------------------------- */
777
778 void fileReport(int type)
779 {
780     Report r = {0};          /* every member starts at zero */
781     const char *detailsPrompt = type == INJURED ? "Describe the injury" :
782                                 type == MISSING ? "Breed, colour, collar and any marks" :
783                                                   "Condition (hungry, friendly, with young...)";
784
785     printf("\n--- Report %s ---\n", type == STRAY ? "a Stray Animal" : type == INJURED ? "an Injured
Animal" : "a Missing Pet");
786     if (type == INJURED)
787         printf("If the animal is bleeding heavily or cannot move, call a vet first.\n");
788
789     readSpecies("Animal (Dog/Cat/Other)    : ", r.species, 1);
790     switch (type) {
791     case STRAY:
792         r.extra.count = readInt("How many animals          : ", 1, 50);
793         break;
794     case INJURED:
795         printf("How serious is it? 1. Urgent  2. Serious  3. Minor\n");
796         switch (readInt("Choose (1-3)              : ", 1, 3)) {
797         case 1: strCopy(r.extra.severity, "Urgent");  break;
798         case 2: strCopy(r.extra.severity, "Serious"); break;
799         default: strCopy(r.extra.severity, "Minor");  break;
800         }
801         break;
802     default:
803         readWords("Pet's name               : ", r.extra.petName, sizeof r.extra.petName);
804         break;
805     }
806     r.date = readDate(type == MISSING ? "Last seen on (dd/mm/yyyy) : " : "Date seen (dd/mm/yyyy)    : ", 0,
0);
807     readText("Area / locality           : ", r.area, sizeof r.area, 2);
808     readWords("City                      : ", r.city, sizeof r.city);
809     readText("Landmark (Enter to skip)  : ", r.landmark, sizeof r.landmark, 0);
810     printf("%s\n", detailsPrompt);
811     readText("  > ", r.details, sizeof r.details, 5);
812     readWords("Your name                 : ", r.reporter, sizeof r.reporter);
813     readPhone("Mobile number             : ", r.phone);
814
815     growReports();
816     r.id = nextReportId++;
817     r.type = type;
818     r.status = OPEN;
819     reports[reportCount++] = r;
820     saveAll();
821     printf("\n  Report #%d submitted. Thank you for helping!\n", r.id);
822 }
823
824 void reportStray(void)   { fileReport(STRAY); }
825 void reportInjured(void) { fileReport(INJURED); }
826 void reportMissing(void) { fileReport(MISSING); }
827
828 void rescueAnimal(void)
829 {
830     const char *OUTCOMES[5] = {"Taken to shelter", "Treated at vet clinic", "Adopted",
831                                "Fed and released", "Returned to owner"};
832     const Report *r;
833     int open = 0, id, idx, i;
834
835     printf("\n--- Rescue a Stray Animal / Mark a Pet Found ---\n");
836     for (r = reports; r < reports + reportCount; r++) {      /* pointer arithmetic */
837         if (r->status != OPEN)
838             continue;
839         if (open++ == 0)
840             printReportHeader();
841         printReport(r);
842     }
843     if (open == 0) {
844         printf("  There are no open reports right now.\n");
845         return;
846     }
847     printLine();
848     id = readInt("Report ID you will take care of: ", 1, 999999);
849     idx = findReport(0, reportCount - 1, id);
850     if (idx < 0 || reports[idx].status != OPEN) {
```

```
851         printf("  ! No open report with ID %d.\n", id);
852         return;
853     }
854     readText("Rescuer / shelter name    : ", reports[idx].rescuer, sizeof reports[idx].rescuer, 3);
855     printf("Outcome:\n");
856     for (i = 0; i < 5; i++)
857         printf("  %d. %s\n", i + 1, OUTCOMES[i]);
858     strCopy(reports[idx].outcome, OUTCOMES[readInt("Choose (1-5)              : ", 1, 5) - 1]);
859     if (!readYesNo("Confirm and save? (y/n)   : ")) {
860         reports[idx].rescuer[0] = reports[idx].outcome[0] = '\0';
861         printf("  Not saved.\n");
862         return;
863     }
864     reports[idx].status = reports[idx].type == MISSING ? FOUND : RESCUED;
865     saveReport(idx);
866     printf("\n  Report #%d marked as %s. Well done!\n", id, STATUS_NAMES[reports[idx].status]);
867 }
868
869 void viewReports(void)
870 {
871     const Report **view;
872     char city[30] = "";
873     int choice, shown = 0, i;
874     int (*compare)(const Report *, const Report *) = byDateNewest;
875
876     if (reportCount == 0) {
877         printf("\n  No reports yet.\n");
878         return;
879     }
880     printf("\nShow: 1. All  2. Open  3. Stray  4. Injured  5. Missing  6. One city  7. All, by city\n");
881     choice = readInt("Choice: ", 1, 7);
882     if (choice == 6)
883         readWords("City: ", city, sizeof city);
884     if (choice == 7)
885         compare = byCity;
886
887     view = calloc(reportCount, sizeof *view);
888     if (view == NULL)
889         return;
890     for (i = 0; i < reportCount; i++)
891         view[i] = reports + i;
892     sortReports(view, reportCount, compare);
893
894     printf("\n");
895     printReportHeader();
896     for (i = 0; i < reportCount; i++) {
897         const Report *r = view[i];
898
899         if ((choice == 2 && r->status != OPEN) ||
900             (choice >= 3 && choice <= 5 && r->type != choice - 2) ||
901             (choice == 6 && !sameText(r->city, city)))
902             continue;
903         printReport(r);
904         shown++;
905     }
906     printLine();
907     printf("%d report(s) shown.\n", shown);
908     free(view);
909 }
910
```

```
Enter your choice: 2

--- Report an Injured Animal ---
If the animal is bleeding heavily or cannot move, call a vet first.
Animal (Dog/Cat/Other)    : Dog
How serious is it? 1. Urgent  2. Serious  3. Minor
Choose (1-3)              : 1
Date seen (dd/mm/yyyy)    : 25/10/2026
Area / locality           : Arundelpet
City                      : Guntur
Landmark (Enter to skip)  : Near school
Describe the injury
  > Hit by a bike, bleeding from the leg
Your name                 : Lakshmi Dheeraja
Mobile number             : 12345
  ! Enter a valid 10-digit mobile number (starts with 6-9).
Mobile number             : 9000012345

  Report #6 submitted. Thank you for helping!
```

Figure 7.1 Reporting an injured animal; a wrong mobile number is rejected.

```
Enter your choice: 1

--- Report a Stray Animal ---
Animal (Dog/Cat/Other)    : cat
How many animals          : 3
Date seen (dd/mm/yyyy)    : 26/10/2026
  ! The date must be today or earlier.
Date seen (dd/mm/yyyy)    : 24/10/2026
Area / locality           : Kothapet
City                      : Guntur
Landmark (Enter to skip)  :
Condition (hungry, friendly, with young...)
  > Three kittens near the bins, hungry
Your name                 : Lakshmi Dheeraja
Mobile number             : 9000012345

  Report #7 submitted. Thank you for helping!
```

Figure 7.2 Reporting a stray animal; a date in the future is rejected and the number of animals is stored in the union.

```
Enter your choice: 3

--- Report a Missing Pet ---
Animal (Dog/Cat/Other)    : Dog
Pet's name                : Tommy
Last seen on (dd/mm/yyyy) : 23/10/2026
Area / locality           : MG Road
City                      : Vijayawada
Landmark (Enter to skip)  : Near temple
Breed, colour, collar and any marks
  > White Pomeranian, blue collar
Your name                 : Lakshmi Dheeraja
Mobile number             : 9000012345

  Report #8 submitted. Thank you for helping!
```

Figure 7.3 Reporting a missing pet; the pet's name is stored in the union.

```
Enter your choice: 4

--- Rescue a Stray Animal / Mark a Pet Found ---
-------------------------------------------------------------------------------------------------
ID  Type     Animal Area           City        Date       Details        Status
-------------------------------------------------------------------------------------------------
1   Injured  Dog    Benz Circle    Vijayawada  24/10/2026 Urgent         Open
    Limping on back left leg, wound on paw
2   Stray    Cat    Brodipet       Guntur      22/10/2026 4 animal(s)    Open
    Mother cat with 3 kittens, looks hungry
3   Missing  Dog    Labbipet       Vijayawada  20/10/2026 Pet: Bruno     Open
    Brown Labrador, red collar
6   Injured  Dog    Arundelpet     Guntur      25/10/2026 Urgent         Open
    Hit by a bike, bleeding from the leg
7   Stray    Cat    Kothapet       Guntur      24/10/2026 3 animal(s)    Open
    Three kittens near the bins, hungry
8   Missing  Dog    MG Road        Vijayawada  23/10/2026 Pet: Tommy     Open
    White Pomeranian, blue collar
-------------------------------------------------------------------------------------------------
Report ID you will take care of: 2
Rescuer / shelter name    : Guntur Animal Shelter
Outcome:
  1. Taken to shelter
  2. Treated at vet clinic
  3. Adopted
  4. Fed and released
  5. Returned to owner
Choose (1-5)              : 1
Confirm and save? (y/n)   : y

  Report #2 marked as Rescued. Well done!
```

Figure 7.4 Rescuing a stray: the open reports are listed and the outcome is saved.

```
911  /* ---------------------------------------------------------------
912   *  My Pet
913   * --------------------------------------------------------------- */
914
915  void listPets(void)
916  {
917      const Pet *p;
918      char dob[16];
919
920      printLine();
921      printf("%-3s %-8s %-6s %-16s %-3s %-11s %-4s %6s  %s\n", "ID", "Name", "Sp.", "Breed", "Sex", "Born",
"Age", "Kg", "Records");
922      printLine();
923      for (p = pets; p < pets + petCount; p++) {
924          formatDate(p->dob, dob);
925          printf("%-3d %-8s %-6s %-16.16s  %c  %-11s %-4d %6.1f  %d\n",
926                 p->id, p->name, p->species, p->breed, p->sex, dob, ageOn(p->dob, today), p->weight, p-
>recordCount);
927      }
928      printLine();
929  }
930
931  void addPet(void)
932  {
933      Pet p = {0};
934
935      printf("\n--- Add a Pet ---\n");
936      readWords("Pet's name                : ", p.name, sizeof p.name);
937      readSpecies("Species (Dog/Cat/Other)   : ", p.species, 1);
938      readText("Breed                     : ", p.breed, sizeof p.breed, 2);
939      for (;;) {
940          char buf[8];
941
942          readLine("Sex (M/F)                 : ", buf, sizeof buf);
943          p.sex = upperCase(buf[0]);
944          if (buf[1] == '\0' && (p.sex == 'M' || p.sex == 'F'))
945              break;
946          printf("  ! Enter M or F.\n");
947      }
948      p.dob = readDate("Date of birth (dd/mm/yyyy): ", 0, 0);
949      p.weight = readFloat("Weight (kg)               : ", 0.1f, 120.0f);
950      readText("Colour / markings         : ", p.colour, sizeof p.colour, 0);
951      for (;;) {
952          int ok, i;
953
954          readText("Microchip (Enter to skip) : ", p.microchip, sizeof p.microchip, 0);
955          ok = p.microchip[0] == '\0' || strLength(p.microchip) == 15;
956          for (i = 0; p.microchip[i] && ok; i++)
957              if (!isDigitChar(p.microchip[i]))
958                  ok = 0;
959          if (ok)
960              break;
961          printf("  ! A microchip number has 15 digits.\n");
962      }
963      growPets();
964      p.id = nextPetId++;
965      pets[petCount++] = p;
966      saveAll();
967      printf("\n  %s added to My Pet with ID %d.\n", p.name, p.id);
968  }
969
970  void showPet(const Pet *p)
971  {
972      char dob[16], d1[16], d2[16];
973      int i;
974
975      formatDate(p->dob, dob);
976      printLine();
977      printf("  %s (ID %d) - %s, %s, %s\n", p->name, p->id, p->species, p->breed, p->sex == 'M' ? "Male" :
"Female");
978      printf("  Born %s (%d years)   Weight %.1f kg   Colour %s\n", dob, ageOn(p->dob, today), p->weight,
979             p->colour[0] ? p->colour : "-");
980      printf("  Microchip: %s\n", p->microchip[0] ? p->microchip : "not chipped");
981      printLine();
982      if (p->recordCount == 0) {
983          printf("  No health records yet.\n");
984          return;
985      }
986      printf("  %-11s %-12s %-30s %s\n", "Date", "Type", "Details", "Next due");
```

```
    for (i = 0; i < p->recordCount; i++) {
        const HealthRecord *h = &p->records[i];

        formatDate(h->date, d1);
        formatDate(h->next, d2);
        printf("  %-11s %-12s %-30.30s %s\n", d1, h->kind, h->detail, d2);
    }
    printLine();
}

int askForPet(void)
{
    int id, idx;

    if (petCount == 0) {
        printf("  No pets yet.\n");
        return -1;
    }
    listPets();
    id = readInt("Pet ID: ", 1, 999999);
    idx = findPet(0, petCount - 1, id);
    if (idx < 0)
        printf("  ! No pet with ID %d.\n", id);
    return idx;
}

void addHealthRecord(void)
{
    const char *KINDS[4] = {"Vaccination", "Deworming", "Check-up", "Treatment"};
    HealthRecord h;
    Pet *p;
    int idx;

    printf("\n--- Add a Health Record ---\n");
    idx = askForPet();
    if (idx < 0)
        return;
    p = &pets[idx];
    if (p->recordCount == MAX_RECORDS) {
        printf("  ! %s already has %d records (the maximum).\n", p->name, MAX_RECORDS);
        return;
    }
    h.date = readDate("Date (dd/mm/yyyy)         : ", 0, 0);
    printf("Type: 1. Vaccination  2. Deworming  3. Check-up  4. Treatment\n");
    strCopy(h.kind, KINDS[readInt("Choose (1-4)              : ", 1, 4) - 1]);
    readText("Details                   : ", h.detail, sizeof h.detail, 3);
    h.next = readDate("Next due (dd/mm/yyyy, 0 if none): ", 1, 1);
    p->records[p->recordCount++] = h;
    saveAll();
    printf("\n  Record added for %s.\n", p->name);
}

void dueReminders(void)
{
    const Pet *p;
    int i, shown = 0;

    printf("\n--- Due in the next 30 days (or overdue) ---\n");
    for (p = pets; p < pets + petCount; p++) {
        for (i = 0; i < p->recordCount; i++) {
            const HealthRecord *h = &p->records[i];
            long days;
            char due[16];

            if (h->next.y == 0)
                continue;
            days = dateToDays(h->next) - dateToDays(today);
            if (days > 30)
                continue;
            formatDate(h->next, due);
            if (days < 0)
                printf("  %-8s %-24s %s  OVERDUE by %ld days\n", p->name, h->detail, due, -days);
            else
                printf("  %-8s %-24s %s  due in %ld days\n", p->name, h->detail, due, days);
            shown++;
        }
    }
    if (shown == 0)
        printf("  Nothing is due in the next 30 days.\n");
```

```
1066  }
1067
1068  void myPetMenu(void)
1069  {
1070      int idx;
1071
1072      printf("\n--- My Pet ---\n");
1073      printf("1. View my pets  2. Pet profile & records  3. Add a pet  4. Add a health record  5.
Reminders\n");
1074      switch (readInt("Choice: ", 1, 5)) {
1075      case 1: printf("\n"); listPets(); break;
1076      case 2:
1077          idx = askForPet();
1078          if (idx >= 0)
1079              showPet(&pets[idx]);
1080          break;
1081      case 3: addPet(); break;
1082      case 4: addHealthRecord(); break;
1083      default: dueReminders(); break;
1084      }
1085  }
1086
```

```
Enter your choice: 6

--- My Pet ---
1. View my pets  2. Pet profile & records  3. Add a pet  4. Add a health record  5. Reminders
Choice: 1

-------------------------------------------------------------------------------------------
ID  Name     Sp.    Breed            Sex Born        Age      Kg  Records
-------------------------------------------------------------------------------------------
1   Coco     Dog    Indie             F  12/06/2022  4      18.0  2
2   Simba    Cat    Persian           M  03/02/2021  5       5.2  1
3   Max      Dog    German Shepherd   M  25/11/2020  5      34.0  1
-------------------------------------------------------------------------------------------

Enter your choice: 6

--- My Pet ---
1. View my pets  2. Pet profile & records  3. Add a pet  4. Add a health record  5. Reminders
Choice: 3

--- Add a Pet ---
Pet's name                : Tiger
Species (Dog/Cat/Other)   : Dog
Breed                     : Beagle
Sex (M/F)                 : M
Date of birth (dd/mm/yyyy): 10/01/2024
Weight (kg)               : 11
Colour / markings         : Tricolour
Microchip (Enter to skip) : 123
  ! A microchip number has 15 digits.
Microchip (Enter to skip) :

  Tiger added to My Pet with ID 4.
```

Figure 8.1 My Pet: the list of pets, then adding a new pet (a wrong microchip number is rejected).

```
Enter your choice: 6

--- My Pet ---
1. View my pets  2. Pet profile & records  3. Add a pet  4. Add a health record  5. Reminders
Choice: 4

--- Add a Health Record ---
-------------------------------------------------------------------------------------------
ID  Name     Sp.    Breed            Sex Born        Age      Kg  Records
-------------------------------------------------------------------------------------------
1   Coco     Dog    Indie             F  12/06/2022  4      18.0  2
2   Simba    Cat    Persian           M  03/02/2021  5       5.2  1
3   Max      Dog    German Shepherd   M  25/11/2020  5      34.0  1
4   Tiger    Dog    Beagle            M  10/01/2024  2      11.0  0
-------------------------------------------------------------------------------------------
Pet ID: 4
Date (dd/mm/yyyy)         : 25/10/2026
Type: 1. Vaccination  2. Deworming  3. Check-up  4. Treatment
Choose (1-4)              : 1
Details                   : Anti-rabies vaccine
Next due (dd/mm/yyyy, 0 if none): 10/11/2026

  Record added for Tiger.
```

Figure 8.2 Adding a vaccination record with its next due date.

```
1087 /* ---------------------------------------------------------------
1088  *  Food & care, Ask LuckyPaws, export
1089  * --------------------------------------------------------------- */
1090
1091 void careGuide(void)
1092 {
1093     char species[8];
1094     int s, stage;
1095     const CareGuide *g;
1096
1097     printf("\n--- Pet Food & Care ---\n");
1098     readSpecies("Species (Dog/Cat)         : ", species, 0);
1099     s = strCompare(species, "Dog") == 0 ? 0 : 1;
1100     printf("Life stage: 1. %s  2. %s  3. %s\n", CARE[s][0].stage, CARE[s][1].stage, CARE[s][2].stage);
1101     stage = readInt("Choose (1-3)              : ", 1, 3) - 1;
1102     g = &CARE[s][stage];
1103
1104     printLine();
1105     printf("  %s - %s\n", species, g->stage);
1106     printLine();
1107     printf("  Meals      : %s\n", g->meals);
1108     printf("  Food       : %s\n", g->food);
1109     printf("  Care       : %s\n", g->care);
1110     printf("  Never feed : %s\n", UNSAFE[s]);
1111     printLine();
1112     printf("  General guidance only - follow your vet's advice.\n");
1113 }
1114
1115 void askLuckyPaws(void)
1116 {
1117     char question[120], lower[120];
1118     int i, k;
1119
1120     printf("\n--- Ask LuckyPaws ---\n");
1121     printf("Ask about food, vaccinations, sickness, toxic foods, strays or missing pets.\n");
1122     printf("Answers come from LuckyPaws' built-in guide - I am not a vet.\n");
1123     printf("Type 'back' to return to the menu.\n");
1124     for (;;) {
1125         readLine("\nYou: ", question, sizeof question);
1126         if (question[0] == '\0')
1127             continue;
1128         toLower(question, lower);
1129         if (strCompare(lower, "back") == 0)
1130             return;
1131         for (i = 0; i < NUM_ANSWERS; i++) {
1132             for (k = 0; ANSWERS[i].keys[k] != NULL; k++)
1133                 if (containsWord(lower, ANSWERS[i].keys[k]))
1134                     break;
1135             if (ANSWERS[i].keys[k] != NULL)
1136                 break;
1137         }
1138         if (i < NUM_ANSWERS)
1139             printf("LuckyPaws: %s\n", ANSWERS[i].answer);
1140         else
1141             printf("LuckyPaws: I don't know that one yet. Try food, vaccination, sick, toxic,\n  stray or
missing - or ask your vet.\n");
1142     }
1143 }
1144
1145 void exportReport(void)
1146 {
1147     FILE *fp = fopen(REPORT_FILE, "w");
1148     const Report *r;
1149     char date[16], extra[40];
1150     int open = 0;
1151
1152     if (fp == NULL) {
1153         printf("  ! Could not create %s\n", REPORT_FILE);
1154         return;
1155     }
1156     formatDate(today, date);
1157     fprintf(fp, "LuckyPaws - Summary Report (%s)\n\n", date);
1158     fprintf(fp, "Rescue & Stray reports\n");
1159     for (r = reports; r < reports + reportCount; r++) {
1160         formatDate(r->date, date);
1161         extraText(r, extra);
1162         fprintf(fp, "  #%-3d %-8s %-6s %-14.14s %-11.11s %s  %-14s %s\n",
1163                 r->id, TYPE_NAMES[r->type], r->species, r->area, r->city, date, extra, STATUS_NAMES[r-
>status]);
```

```
1164          if (r->status == OPEN)
1165              open++;
1166      }
1167      fprintf(fp, "\nPets: %d   Open reports: %d   Animals helped: %d\n", petCount, open, reportCount -
open);
1168      fclose(fp);
1169      printf("\n  Summary written to %s\n", REPORT_FILE);
1170  }
1171
```

```
Enter your choice: 7

--- Pet Food & Care ---
Species (Dog/Cat)        : cat
Life stage: 1. Kitten (under 1 year)  2. Adult (1-10 years)  3. Senior (11+ years)
Choose (1-3)             : 1
-----------------------------------------------------------------------------------------
  Cat - Kitten (under 1 year)
-----------------------------------------------------------------------------------------
  Meals      : 3-4 small meals a day
  Food       : Kitten food (wet and dry); fresh water always
  Care       : Vaccination course from 8 weeks as the vet advises; deworm as advised; litter training
  Never feed : lilies (all parts), onion and garlic, chocolate, grapes and raisins, cow's milk, dog food as a diet
-----------------------------------------------------------------------------------------
  General guidance only - follow your vet's advice.
```

Figure 9.1 Food & care guide for a kitten.

```
Enter your choice: 8

--- Ask LuckyPaws ---
Ask about food, vaccinations, sickness, toxic foods, strays or missing pets.
Answers come from LuckyPaws' built-in guide - I am not a vet.
Type 'back' to return to the menu.

You: Is chocolate bad for dogs?
LuckyPaws: Never give chocolate, grapes, onion, garlic or xylitol. Lilies are very dangerous
  for cats. If eaten, contact a vet immediately. See menu option 7.

You: My dog has a fever
LuckyPaws: A pet that stops eating, keeps vomiting or is very weak should see a vet soon.
  Never give human medicines - many are poisonous to pets.

You: How do I make tea?
LuckyPaws: I don't know that one yet. Try food, vaccination, sick, toxic,
  stray or missing - or ask your vet.

You: back
```

Figure 9.2 Ask LuckyPaws: answers found by keyword, and a reply when the question is not known.

```
Enter your choice: 9

  Summary written to luckypaws_report.txt

Enter your choice: 0

Data saved to luckypaws.dat. Thank you for caring for animals!

C:\Users\golth\OneDrive\Documents\LuckyPaws>type luckypaws_report.txt
LuckyPaws - Summary Report (25/10/2026)

Rescue & Stray reports
  #1   Injured  Dog    Benz Circle    Vijayawada  24/10/2026  Urgent         Open
  #2   Stray    Cat    Brodipet       Guntur      22/10/2026  4 animal(s)    Rescued
  #3   Missing  Dog    Labbipet       Vijayawada  20/10/2026  Pet: Bruno     Open
  #4   Stray    Dog    Thullur Road   Amaravati   18/10/2026  1 animal(s)    Rescued
  #5   Injured  Cat    Kothapet       Guntur      15/10/2026  Serious        Rescued
  #6   Injured  Dog    Arundelpet     Guntur      25/10/2026  Urgent         Open
  #7   Stray    Cat    Kothapet       Guntur      24/10/2026  3 animal(s)    Open
  #8   Missing  Dog    MG Road        Vijayawada  23/10/2026  Pet: Tommy     Open

Pets: 4   Open reports: 5   Animals helped: 3

C:\Users\golth\OneDrive\Documents\LuckyPaws>
```

Figure 9.3 Exporting the summary report, saving and exiting, then viewing the file with the type command.